

09/28/2025: Partitions

CSCI 246: Discrete Structures

Textbook reference: Sec 16, Scheinerman

Today's Agenda

- Review early course feedback (≈ 10 mins)
- Mini-lecture (≈ 25 mins)
- Group exercises (≈ 15 mins)

Early course feedback

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Partitions: the intuition

From equivalence relations to partitions

From partitions to equivalence relations

Using equivalence relations to count

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Example (continued). Let A be the students, TA, and teacher for the class. We've formed the partition $A = A_0 \cup A_1 \cup A_{2+}$ where individuals in A_i have i siblings:

$$A_0 = \{\text{???}, \dots\}$$

$$A_1 = \{\text{Mike W., Madhav}, \dots\}$$

$$A_{2+} = \{\text{???}, \dots\}$$

Partitions in everyday life

Sorting laundry. Whites, darks, colors. Every item goes in one basket, no item is in two, and nothing is left on the floor.



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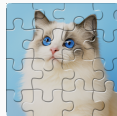
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Poll. Any other examples?

Another demo. Stand by the whiteboard that best characterizes you:
do you have a dog or a cat?

Partition failures

Another demo. Stand by the whiteboard that best characterizes you:
do you have a dog or a cat?

Assessment. This demo breaks! Someone has both, someone has
neither (or a goldfish). We can't form a partition of the class this way.

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Poll. Is the following an equivalence relation on the set $A = \{1, 2, 3, 4, 5, 6\}$?

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), \\ (4, 4), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)\}$$

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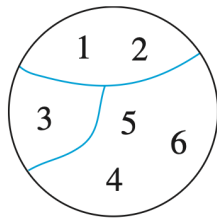
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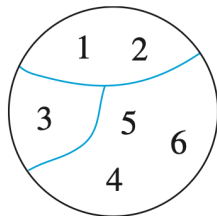
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Reminder. The notation $[a]$ is the equivalence class associated with a .

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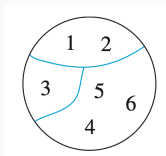
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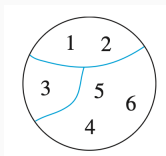
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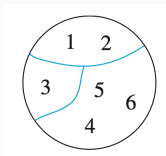
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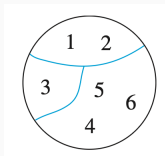
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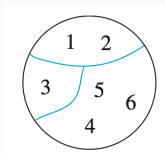
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From equivalence relations to partitions: The general case

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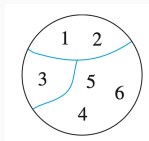
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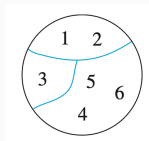
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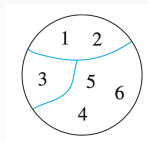
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This phenomenon is true **generally**.

Cor. 15.13 Scheinerman (restated on pp. 85)

Let R be an equivalence relation on a set A . The equivalence classes of R form a partition on the set A .

From equivalence relations to partitions: Another example

Proposition (Scheinerman pp. 85)

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Example

Let's set R to congruence mod 3. We've shown R is an equivalence relation. Thus, we can partition $A = \{\text{the integers}\}$ into the following equivalence classes

$$[0] = \{\dots, -6, -3, 0, 3, 6, \dots\}$$

$$[1] = \{\dots, -5, -2, 1, 4, 7, \dots\}$$

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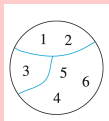
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From partitions to equivalence relations

Example

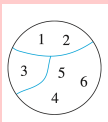


Starting from the partition on the left, we can reconstruct the equivalence relation:

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), \\ (4, 4), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)\}$$

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Definition

Let $\mathcal{P}$ be a partition on a set A . The **is-in-the-same-part-as** relation, denoted $\stackrel{\mathcal{P}}{=}$, is given by

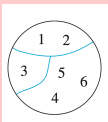
$$a \stackrel{\mathcal{P}}{=} b \iff \exists P \in \mathcal{P} : a, b \in P$$

Proposition (Scheinerman Prop 16.3)

Let $\mathcal{P}$ be a partition on a set A . The relation $\stackrel{\mathcal{P}}{=}$ is an equivalence relation on A .

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Remark

In gist, we can take elements in the same cell to be related to each other.

Partitions: the intuition

From equivalence relations to partitions

From partitions to equivalence relations

Using equivalence relations to count

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Question

How many ways are there to rearrange the letters in the word HELLO?

Using equivalence relations to count

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Solution

If there were no repeated letters (imagine a big and small L), the answer would be $5! = 120$.

HELLO	HELOL	HELLO	HELLOL	HEOLL	HEOLL
HLELO	HLEOL	HLLEO	HLLOE	HLOLE	HLOEL
		<i>many lines omitted</i>			
LLHEO	LLHOE	LLEHO	LLEOH	LLOHE	LLOEH
LLHEO	LLHOE	LLEHO	LLEOH	LLOHE	LLOEH

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But we want to treat the big and small L as equivalent. So we form equivalence classes, each of which have 2 members.

$$[\text{HLEOL}] = \{\text{HLEOL}, \text{HLEOL}\}$$

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All together there are $\frac{120}{2} = 60$ equivalence classes. Hence, the answer is 60.

Theorem: Counting Equivalence Classes (Scheinerman Thm 16.6)

Let R be an equivalence relation on a finite set A . If all the equivalence classes of R have the same size, m , then the number of equivalence classes is $|A|/m$.

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Remark. You'll practice applying this theorem during the group exercises.